

Documentation: Hall A Moeller Quad Power Supplies

4/5/2000

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Major components:

- Sorensen DHP40-330 switching power supplies
- High current polarity reversing switch
- Automation Direct series D2 PLC for interlock and controls

Interface and Controls:

- CAMAC DAC, ADC, digital I/O controlled by EPICS application software
- Digital bit control for status and control
- 0-10 VDC analog for current programming and readback
- Local PLC for interlocks & polarity control

Capabilities:

- 0-40 VDC (bipolar using reversing switch)
- 0-330 A DC
- Additional specifications per Sorensen data sheet

Hardware

Three identical channels are provided for control of magnets MQM1H02, MQO1H03, MQO1H03A. Each channel consists of a Sorensen DHP series power supply capable of supplying 330 A DC at up to 40 V. The output of each power supply feeds to a high current, motorized reversing switch. The switch is controlled via a PLC located in the same rack as the three power supplies and three reversing switches.

The Sorensen DHP series features a digital front panel and combination digital and analog remote interface. In operation, the desired output current is set from the CAMAC DAC channel associated with the specific power supply. The higher level software (EPICS) controls ramping, hysteresis, and selects reversing switch polarity. The local PLC provides the interface between CAMAC and the reversing switches, power supplies, and external interlocks.

Equipment overview.

Sorensen DHP power supply.

This series is available in a wide range of voltages, currents, and powers. As it is a modular design, higher current/power models are realized by using multiple output modules. The DHP40-330 contains four such output blocks contained within a single 10.5" (6U) rack mount chassis. Front panel displays show voltage and current settings, plus mode and status signals. Front panel controls include on/off, function keys, and a numeric keypad.

The power supply is powered from a 208 V/ 3 phase source of approximately 50 A at maximum output. A connectorized pigtail attached to the power supply permits reasonably quick replacement should replacement be required. Remote control is accomplished through a single chassis mounted D25 connector (female) on the rear of the power supply. DC output connections are via two solid bus bars at the rear of the power supply.

The power supply may be operated in constant voltage, constant current, or constant power modes. As configured (for remote operation), the power supply is operated in constant current mode. Desired output current is programmed remotely using a 12 bit CAMAC DAC channel via an EPICS operator interface screen. Readback is via the 0-10 VDC output from the Sorensen. **NOTE: This output provides only 8-bit resolution at this time. Sorensen will be releasing a new control board that will provide higher resolution sometime in the early summer of 2000. The power supplies will be upgraded when this new board becomes available.**
DHP40-330 Configuration.

The power supply must be configured for remote operation using the DHP40-330 front panel controls.
Failure to do this may result in loss of remote control or improper operation!

When power is turned on, the power supply will go through a series of self-tests. Once these tests have been completed and a static message is displayed on the front panel, the power supply may be programmed. The selections you make are automatically stored in non-volatile memory and *will not be lost* should the supply lose power. (Refer to Section 3.2 of the Sorensen DHP Manual 11-195-000-00 REV D for complete programming options.)

1. **CONTROL SOURCE.** Cycle the MENU and scroll to this function. Use the UP/DN arrow keys to scroll through the choices. Select "REM ANALOG" option when it is displayed by pressing ENTER.
2. **ANALOG CONTROL OPTION.** Press MENU and scroll to this function. Select "10V & CURRENT". Press ENTER to store this selection.
3. **SET VOLTAGE LIMIT.** Press VOLTAGE, key in "30" and press ENTER to store the value. Note that this value can be anything above the voltage required to achieve the desired output current. Setting it to 30 means that if a current has been pre-programmed and the power supply is switched ON, the output voltage will rise to 30V trying to get the current into the magnet as fast as possible.
4. **SET OVERVOLTAGE.** Press OVERVOLTAGE button, key in "44" and press ENTER to store the value. **IMPORTANT:** if the overvoltage value is set to a voltage that will be reached, the power supply will trip off. Since the maximum output is 40 V, setting the trip to 44 will insure that the supply will trip only in the event of a real fault. This supply should NEVER be capable of outputting 44 Volts.

Reversing Switch.

As the Sorensen is a unipolar power supply, bipolar operation is obtained by using a polarity reversing switch. The switch, a Kraus & Naimer series L351 supplied by American Solenoid (732-560-1240) is rated for full voltage and 350 A. It is a mechanical design equipped with a motor driven selector and auxiliary contacts that report the present position of the switch.

Switch position and motion is controlled by the PLC and EPICS-CAMAC application software. In operation, EPICS requests a polarity. The PLC will process this request when the output current has been reduced to within a few amps of zero. The switch will change polarity (if requested by EPICS) as current ramps through zero.

PLC.

The PLC (Programmable Logic Controller) is modular industrial system purchased from Automation Direct. It consists of a six slot I/O base with power supply, D2-240 CPU, one 4-channel 12-bit ADC card (model F2-04AD-2), two 16-point DC input modules (model D2-16ND3-2 ; one modified for low voltage control), one 12-point relay module (model D2-12TR), and one 4-point relay module (model D2-04TR).

The PLC interfaces between CAMAC, the DHP40-330s, the reversing switches, and interlocks (magnet current, magnet temperature fault Klaxons and rear door interlocks). The magnet current interlock permits the polarity switch to change states only when magnet current is near zero. This insures long contact life by preventing high currents, and resulting high back EMFs, from being switched.

How it works. A request for a polarity change is issued by the EPICS software through CAMAC and to the PLC. This normally happens as the value of current ramps to zero and changes polarity. When the current is low enough (about a 5 A window around zero) the PLC will activate the switch motor and change to the requested polarity. The polarity state is then reported back to the EPICS software, which determines the polarity of the magnet current and displays the appropriate sign.

The PLC's program is contained in non-volatile memory on the PLC CPU card. Programming is entered in ladder logic on a PC, and then downloaded to the PLC via its RS-232 port. The D2-240 has two RS-232 ports. The primary port is connected to a PC for programming or monitoring. The secondary port can be connected to a local user interface panel (in this case containing 4 pushbuttons, 4 LEDs, and two 20 character LCD lines). The local interface panel displays any faults, present polarity status, and allows the switch position to be

changed when local control mode has been selected. NOTE: the "RUN/Term" switch on the CPU card must be in "Term" position for the buttons on the operator interface to function.

Diagnostics. Each module has a multi-LED display. These LEDs indicate the present status of inputs and outputs. Because the LEDs always display the present status of the signals associated with them, they are perfect diagnostic indicators. Each interlock, status bit, or control bit has an LED associated with it. By referring to the input/output assignment sheet, the present status of each can be easily determined. Note that some modules on this PLC only display HALF of the signals on that module at any given time. A selector switch on the front of the module allows selection of either the upper or lower half for display.

Control Power.

A separate 24 VDC, 1.2A power supply is affixed to the rear of the display panel just below the PLC (on the local control panel. This supply provides power for all 24 V I/O (switches, interlocks, CAMAC controls, etc.). Source of power supply: AutomationDirect.

CAMAC Interface.

Three cards are required for this installation. A Jorway IO card is used for digital IO (control and status bits), a Bira 5333B ADC card (12-bit with three 0-10 V channels used), and a Bira 5408 12-bit DAC (with three 0-10V channels used).

EPICS controls.

The interface screen is the same as was used with the Lakeshore power supplies. In operation a current is set. The code will send out values to the DAC (used to set the power supply current) in increments defined by the column labeled "Amps/sec ramp rate". Note that rather than a continuous ramping, the output current will actually increase in discrete steps of the "Amps/second" value.

The magnet current readback is derived from a 0-10 V signal from each power supply as applied to a Bira 5333B ADC card. Sign (polarity) is determined by reading the auxiliary status contact through CAMAC and the PLC. Unfortunately, while the power supply is settable to 12-bit accuracy, the readback is only 8 bits. For this 330A power supply, that means a jitter or uncertainty of even a single bit will result in the displayed current to flicker by over 1A.

Operator Interface Panel (Optimation model OP-420).

This control panel (located on the panel directly below the PLC) consists of 4 pushbutton switches with status LEDs, plus a two-line LCD display. The LCD display shows faults and the present polarity setting of each of the reversing switches. The button labeled LOCAL-REMOTE selects the control point for the three power supplies and reversing switches. ***For normal remote operation this switch must be set to REMOTE and the LED on this button will not be lit.*** When set to REMOTE, the other three pushbuttons will have no effect.

Toggling the LOCAL-REMOTE button will allow/disallow two fundamental remote functions: ON/OFF and POLARITY SELECT. Output current continues to be controlled by the CAMAC DAC setting until the power supply has been reprogrammed using the Sorensen front panel keypad. Unless faults are present, all three Sorensen supplies will be enabled (turned on), and the output current will be set to the present setting (either from the CAMAC DAC or the Sorensen front panel depending on which source has been selected). NOTE: Output polarity can not be changed without first setting output current to zero. This means the polarity pushbuttons will not change the output polarity unless current is first set to zero.

The primary use for LOCAL mode control is only for ***testing and diagnostics***. All three power supplies are enabled when in LOCAL mode. Individual power supplies may be turned off using their front panel power switches.

Hardware Installation.

A single rack is holds the three power supplies, three switches, PLC, display panel, and control power. Separate cables are wired to the PLC interface panel (euro-style terminal strips). The cables connect to: CAMAC DAC, CAMAC ADC, CAMAC I/O, the three switches, and the three Sorensen power supplies. The other ends of all but the switch cables are also connectorized: D25M for the power supplies, and appropriate connectors for the various CAMAC cards.

AC power is connections are made by 3 twistlock connectors (the Sorensen power supplies) plus two standard 120 V line cords.

Maintenance considerations.

Spares. A complete set of spare PLC modules was purchased. As all modules are connectorized (both to the base and to the I/O connections), a bad module may be easily replaced. The spare CPU module may be preprogrammed for quick replacement or logic changes as well.

Sorensen power supply replacement. Lock and tag in accordance with JLab requirements. Each power supply is capable of supplying 330 amps at 40 volts and represents a **Class 2** electrical hazard. Due to the close proximity of other identical power supplies, it may be appropriate to *lock out all three Sorensen power supplies* when performing work on any power supply.

Unbolt the output leads from the power supply. If the replacement power supply does not already have AC power leads attached, disconnect the AC leads from the supply being removed and reinstall them on the replacement supply. Unplug the DB-25 interface connector (J1). The supply may then be unbolted from the rack and the replacement installed.

PLC replacements will typically require a single card to be exchanged. As mentioned previously, these modules are fully connectorized. Unplug the IO connections from the card in question, lift the locking tabs and remove the card. Note that ONE of the input modules has been specially modified to work with the output bits from the Sorensen power supplies. It **MUST** be replaced with a similar module should replacement be required.

Switch replacement is the most complex due to the size of the DC cables attached. Removal and reinstallation should be straightforward.

Attachments:

System block diagram

Cable Information

Connection/Connector pinout information

Terminal block assignments

Sorensen User's manual and other available service info (these supplies have come with a 5 year warranty)

Manuals: Sorensen, this document, PLC and Optimate devices & software

Switch data

LEM current sensor data (if this device was used)

CAMAC slot & card types and assignments

PLC card placement

PLC IO assignments

PLC code (printout & CD)

PLC programming software (on CD, diskette available)

Operator interface panel code (printout & CD)

Operator interface programming software (Optimation) (on CD)